EE2200 - Electrical Machines (Semester: Fall 2026)
Quiz 1 : DC Machines, AC Machines and Transformers

Name: Roll no: Code : YYWMGF

Scan / click for YouTube playlist on DC Machines.

Scan / click for YouTube playlist on Induction Machines.

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1 DC Machines

1. A 8 pole DC shunt generator with 20 wave connected armature conductors is running at 540 rpm and supplying a load of 80 Ω resistance at 220 V terminal voltage. The armature resistance is 0.2 Ω and field resistance is 200 Ω. Find the emf generated in the armature.

   A. 220.770 V

   B. 225.910 V

   C. 221.820 V

   D. 239.550 V

2. A DC generator has armature emf of 170 V when flux per pole is 10 mWb, and speed is 1250 rpm. Calculate the emf generated when speed is 1260 rpm and flux per pole is 26 mWb.

   A. 405.438 V

   B. 504.152 V

   C. 445.536 V

   D. 434.936 V

3. A DC shunt generator delivers 50 kW at 240 V and 330 rpm. The armature and field resistance are 0.027 Ω and 50 Ω respectively. The speed of the machine when running as a shunt motor and taking 50 kW input at 240 V will be

   A. 332.093 rpm

   B. 322.643 rpm

   C. 314.893 rpm

   D. 337.893 rpm

4. A 15 kW 300 V short shunt compound DC generator has a full load efficiency of 69.0 %. If the resistance of armature and interpole is 0.4 Ω and that of series and shunt winding are 0.65 Ω and 250 Ω respectively. Calculate the combined bearing friction loss, windage and core losses of the machine.

   A. 3609 W

   B. 3292 W

   C. 3618 W

   D. 572 W

5. A belt driven DC shunt generator runs at 1330 rpm delivering 9 kW, at 210 V bus. The belt breaks following which the machine operates as a motor drawing 3.7 kW power. The armature and field resistance are 0.39 Ω and 56 Ω respectively. The speed when working as a motor will be

   A. 1192.52 rpm

   B. 1064.52 rpm

   C. 962.52 rpm

   D. 1470.52 rpm

6. A DC shunt motor supplied at 275 V, runs at 1490 rpm and the armature current drawn is 44 A. The resistance of the armature circuit is 0.4 Ohm. The resistance required to be added in series with the armature to readuce the speed to 700 rpm when the armature current is 21 A, is ?

   A. 0.438

   B. 3.102

   C. 7.337

   D. 6.937

7. A 355 V DC shunt motor draws a line current of 5.4 A, on no load. If armature resistanc is 0.35 Ω and field resistance is 280 Ω, determine the efficiency of the machine running as a generator delivering a load current of 36 A at 355 V.

   A. 84.206 %

   B. 80.423 %

   C. 75.789 %

   D. 90.017 %

8. A 2 pole wave wound motor armature has 550 conductors and delivers 155 A. The brushes have been displaced through 3.05 angular degrees from geometrical axis. Find additional field current for neutralising demagnetising effect if there are 1130 turns in shunt field winding.

   A. 0.470 A

   B. 0.291 A

   C. 0.268 A

   D. 0.320 A

9. A 200 V DC shunt motor takes 2 A at no load. It draws 28 A when running at full load at 1540 rpm. The armature and shunt resistance are 0.53 Ω and 200 Ω, respectively. The no load speed in rpm is

   A. 1628.28 rpm

   B. 1570.78 rpm

   C. 1679.88 rpm

   D. 1654.28 rpm

10. A 205 V DC shunt motor has an armature resistance of 0.5 Ω and field resistance of 55 Ω. When the motor is operated on no-load at rated voltage it drawns an armature current of 4 A and runs at 1310 rpm. Now the machanical load on the motor is gradually increased, until the total line current drawn by the motor reaches 32 A at rated voltage. It was noted that the air-gap flux has been reduced by 6 % due to armature reaction. Voltage drop across brushes can be taken as 1.5 V per brush under all operating conditions. Calculate the speed of the motor under the above loaded condition.

    A. 1444 rpm

    B. 1309 rpm

    C. 1231 rpm

    D. 1495 rpm

11. A separately excited DC motor has an armature resistance of 0.6 Ω. It runs from a 255 V DC supply drawing an armature current 19 A at 1270 rpm. For the same field current, the torque developed (in Nm) for an armature current 8 A will be

    A. 12.48 Nm

    B. 13.77 Nm

    C. 14.66 Nm

    D. 17.12 Nm

12. A long shunt compound DC generator delivers a load of 95 A at 220 V. The armature, series and shunt winding resistances are 0.087 Ω, 0.067 Ω and 75 Ω respectively. Determine the EMF generated. Allow a brush contact drop of 1 V per brush.

    A. 237.082 V

    B. 192.036 V

    C. 260.582 V

    D. 232.630 V

13. At 1155 rpm the induced emf of a DC machine is 138 V and the armature current is 41 A. The electromagnetic torque produced would be?

    A. 46.78 Nm

    B. 38.04 Nm

    C. 52.96 Nm

    D. 41.31 Nm

14. A break test is conducted on a DC motor. The spring balance reading are 4 kg and 32 kg. The diameter of drum is 48 cm. The speed of the motor under these loading is 750 rpm, when supplied at 220 V, taking current of 30 A from the mains. Find efficiency.

    A. 77.685 %

    B. 76.345 %

    C. 68.575 %

    D. 78.425 %

15. A 42.0 kW, 420 V, 4 pole wave wound DC shunt motor has 732 armature conductors and 170 commutator segments. Its full load efficiency is 82.0 % and the shunt field current is 1.0 A. If the brushes are shifted backward by 1.9 segments from geometrically neutral axis. Find the demagnetising ampere turns per pole.

    A. 398.283 AT/pole

    B. 462.562 AT/pole

    C. 496.142 AT/pole

    D. 494.762 AT/pole

16. A 8 pole DC generator has induced emf of 200 V when driven at a speed of 170 rpm.The armature is lap wound and has 30 conductors. The armature resistance is 0.25 Ω. The pole shoe has diameter of 20 cm, each pole substends an angle of 35 degrees at the center. The length of pole is 35 cm. Determine the flux density in the air gap.

    A. 0.360 Wb/m^2

    B. 0.450 Wb/m^2

    C. 0.610 Wb/m^2

    D. 0.110 Wb/m^2

Sir in mail informed that options should be multiplied by 100

17. A 8 pole lap connected DC machine has an armature resistance of 0.25 Ω, and generate 90 volt on no load at rated speed. Determine the armature resistance, when same machine is re-wound using simplex wave connection.

    A. 4.000 Ω

    B. 8.000 Ω

    C. 3.000 Ω

    D. 2.060 Ω

18. A 2 pole DC generator, having wave wound armature winding has 40 slots, and 45 conductors per slot. Flux per pole is 19 mWb. What is the emf generated when motor is driven at 180 rpm.

    A. 107.730 V

    B. 106.600 V

C. 102.400 V

D. 102.600 V

19. Consider a saperately excited DC motor rated at 150 V, 25 A. It has an armature resistance of 0.6 Ohms. The rated DC voltage is applied to both the armature and field windings. The motor runs at a speed of 1100 rpm, while its armature draws a current of 5.8 A from the source, the field winding draws a current of 2.1 A. The torque developed by the motor is?

    A. 9.06 Nm

    B. 6.08 Nm

    C. 7.38 Nm

    D. 8.52 Nm

20. A saperately excited DC motor rated 281 V, 20 A, 1650 rpm drives a constant torque load at rated speed operating from rated supply drawing rated current. The armature resistance is 0.7 Ω. If the supply voltage drops by 11 % with filed current unaltered, then resultant speed of the motor in rpm is?

    A. 1459 rpm

    B. 1276 rpm

    C. 1748 rpm

    D. 1650 rpm

21. A 33.8 kW, 300 V, 6 pole wave wound DC shunt motor has 795 armature conductors and 120 commutator segments. Its full load efficiency is 80.0 % and the shunt field current is 3.1 A. If the brushes are shifted backward by 2 segments from geometrically neutral axis. Find the distorting ampere turns per pole.

    A. 3467 AT/pole

    B. 3650 AT/pole

    C. 912 AT/pole

    D. 8212 AT/pole

22. A 8 kW, 240 V, DC shunt motor with an armature resistance of 0.4 Ω and field resistance 290 Ω takes 3.5 A, when running light at rated voltage and rated speed. What is machine efficiency as motor with input of 8 kW at rated speed and voltage.

    A. 84.253 %

    B. 85.358 %

    C. 91.800 %

    D. 86.153 %

23. A 200 V, 11 kW, 940 rpm separately excited DC motor has an armature resistance 0.022 Ω. When the motor operates at rated speed and with rated terminal voltage , the electromagnetic torque developed by the motor is 70 Nm. Neglecting the rotational losses of the machine, the current drawn by the motor from 200 V supply is

    A. 37.89 A

    B. 34.57 A

    C. 32.69 A

    D. 33.76 A

24. A 286 V, 11.5 kW, 1430 rpm saperately excited DC motor has in armature resistance of 0.05 Ohms. When the motor operates at rated speed and with rated terminal voltage, the electro magnetic torque developed by the motor is 53 Nm. Neglecting the rotational losses of the machine, the armature current drawn by the motor from the supply is?

    A. 14.64 A

    B. 5.69 A

    C. 23.898 A

    D. 27.888 A

25. The resistance measured acorss the terminals of a DC series motor was found to be 0.45 Ohm. When a voltage of 375 is applied acorss the terminals, the motor was taking 32 A and running a load at 1170 rpm. If an additional resistance of 5.5 ohm is added in series to the armature, and if the armature current remains same, what will be the speed of the motor? (Hint: The terminal resistance of DC series motor includes resistance of both armature and field.)

    A. 599 rpm

B. 514 rpm

C. 519 rpm

D. 537 rpm

26. A 5 kW, 180 V DC shunt motor has armature resistance of 0.97 Ω and shunt field resistance 90 Ω. At no load, the motor draws 7 A from 180 V supply and runs at 1119 rpm. The rotational loss of the machine is

A. 1077.8 W

B. 681.9 W

C. 384.2 W

D. 875.8 W

27. A separately excited generator when running at 1080 rpm supplieS 60 A to a resistive load at 220 V. What will be the load current when the speed drops to 560 rpm, if field current is unchanged. Given armature resistance is 0.5 Ω.

A. 31.111 A

B. 29.711 A

C. 36.611 A

D. 33.911 A

2 Induction Motors

28. A 3-phase, 200 V, 60 Hz, 4 pole induction motor runs at 1620 rpm at full load. Calculate the full load slip of the motor.

A. -0.100

B. 0.100

C. 0.111

D. 0.120

29. A 3-phase 40 Hz, 10 pole induction motor has a rotor resistance of 0.04 Ω and a standstill rotor reactance of 1.1 Ω. Neglect the voltage drop in stator and assume that the rotor resistance remains constant. Given that the full load slip is 7 %, the ratio of maximum torque to full load torque is?

A. 1.4881

B. 1.5257

C. 1.2223

D. 1.038

30. A 3-phase, 50 Hz, 8 Pole induction motor has a rotor impedance of (0.03 + j0.15) ohm at stand still. If the full load torque is obtained at 682.5 rpm, Calculate the ratio of starting torque to full load torque

A. 0.514

B. 0.534

C. 0.257

D. 0.494

31. In a 3-phase, 40 Hz, 6 Pole induction motor, the rotor electromotive force is observed to make 3.60 complete alterations per second. Find the ratio of rotor copper loss to the mechanical power developed in the rotor. Calculate upto 4 decimal places.

A. 0.1978

B. 0.0989

C. 0.3293

D. 0.1098

32. The power input to the rotor of a 3-phase, 40 Hz, 6 pole induction motor is 17.0 kW, the rotor copper losses are 340.0 **W per phase**. Calculate the speed of rotor in rpm, ignoring the mechanical losses.

A. 706.9

B. 800.0

C. 752.0

D. 48.0

33. A 3-phase, 6 pole, 210V, 55 Hz induction motor has a full load slip of 9 %. Calculate the frequency of emf induced in the rotor at full load in Hz.

A. 50.05

B. 27.50

C. 4.95

D. 110.00

34. A 18 pole 3-phase induction machine is working as an induction generator. The supply frequence is 85 Hz. While the frequency of the rotor current is 7.5 Hz. The mechanical speed of the rotor in rpm is?

    A. 671 rpm

    B. 617 rpm

    C. 517 rpm

    D. None of the options

35. A 3-phase, 220 V, 45 Hz, 4 pole induction motor has a full load slip of 0.06. Calculate the rotor speed of the motor in rpm.

    A. 76

    B. 81

    C. 1193

    D. 1269

36. A 3 ϕ induction motor is wound for 6 poles and is supplied from 45 Hz system. Calculate the synchronous speed in rpm.

    A. 1800

    B. 900

    C. 910

    D. 450

37. An 12 pole, 45 Hz, 3-phase induction motor has blocked rotor reactance per phase which is 6.7 times the rotor resistance per phase. The speed at which maximum torque developes is ... ?

    A. 383 rpm

    B. 393 rpm

    C. 394 rpm

    D. 450 rpm

38. A 3-phase 45 Hz, 10 pole induction motor runs at 476 rpm. The stator copper loss, core loss, and rotational losses of motor can be neglected. The percentage efficiency of motor is close to

    A. 88.2 %

    B. 74.2 %

    C. 86.7 %

    D. 90.4 %

39. A 3-phase squirrel cage induction motor has starting torque of 160 % and maximum torque of 360 % with respected to the rated full load torque at rated voltage and rated frequency. Neglect stator resistance and rotational losses. The value of slip for maximum torque is?

    A. 19.63 %

    B. 23.44 %

    C. 19.06 %

    D. None of the Options

40. The power input to a 420 V, 45 Hz, 12 Pole 3-phase induction motor running at 415 rpm is 82 kW. The stator copper losses and stator core losses togeather totals to 3 kW. The friction and windage losses total up to 1 kW. What is the efficiency of the motor?

    A. 88.38 %

    B. 88.96 %

    C. 87.61 %

    D. 85.68 %

41. The stator copper loss of a 3-phase induction motor is 3.8 kW. Taking 55 kW from the supply it runs at a slip of 3.2 %. The losses in the stator core can be assumed as 1.6 kW. Calculate the mechanical power developed by the motor.

    A. 39.99 kW

    B. 51.322 kW

    C. 48.013 kW

    D. 52.749 kW

42. A 3-phase, 720 kW, 50 Hz, 8 Pole induction motor has a rotor impedance (0.023 + j0.06) ohm at standstill. Calculate the rotor resistance in Ω to be added to get maximum starting torque.

    A. 0.083

    B. 0.037

C. 0.249

D. 0.111

43. A 3-phase, 230 V, 40 Hz induction motor runs has a synchronous speed of 600.0 rpm. Calculate number of poles of the motor.

A. 8

B. 16

C. 9

D. 4

44. A 746 kW, 3 phase, 45 Hz, 8 pole induction motor has rotor resistance and reactance at stand still of 0.07 Ω and 0.28 Ω per phase. The full load torque is obtained at 648.0. Find the ratio of maximum torque to full load torque.

A. 3.205

B. 3.225

C. 1.603

D. 3.185

45. A 410 V, 40Hz, 30 HP 3-phase induction motor is drawing 76 A current at 0.88 power factor lagging. The stator and rotor copper losses are 1.1 kW and 1.2 kW respectively. The friction and windage losses are 0.65 kW and the stator core losses are 1.7 kW. The air gap power of the motor is?

A. 45.194 kW

B. 44.494 kW

C. 44.694 kW

D. 43.494 kW

3 Transformers

46. A 4 kV/400 V, 400 kVA transformer has a primary winding resistance of 0.5 Ω and a secondary winding resistance of 0.003 Ω. The iron loss is 3.75 kW. If the power factor of the load is 0.85 lag, determine the efficiency of the transformer on full load.

A. 93.43 %

B. 96.66 %

C. 93.66 %

D. 97.16 %

47. A 320/160 V transformer has an iron loss of 805 W when excited from high voltage side under no load condition. If the no load current is 7.86 A, then the no load power factor is

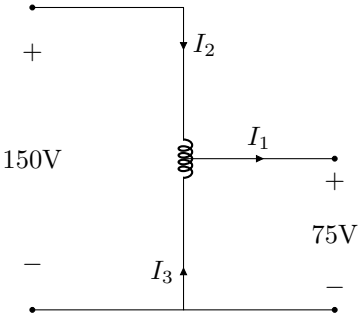
A. 0.52 lag

B. 0.68 lag

C. 0.59 lag

D. 0.32 lag

48. A single-phase auto-transformer has a voltage ratio 150/75 V and supplies a load of 20 kVA at 75 V. Assuming an ideal transformer, determine the current I_3 in the common section of the winding as shown in the circuit diagram.



A. 133.33 A

B. 133.58 A

C. 136.67 A

D. 134.33 A

49. An ideal transformer has a turns ratio of 11:130. The following connections are made:
Primary side: Voltage of 200 V is applied
Secondary side: a resistance of 9 kΩ is connected.
What is the input power taken by the primary side?

A. 584 W

B. 590 W

C. 621 W

D. 53 W

EMF Equation of Transformer

A 27 KVA transformer has 380 turns in the primary and 50 turns in the secondary. The primary is connected to 2790 V, 40Hz supply. Neglect losses, leakage drops and no load current in the primary. Based on this data solve following three questions.

Hint: Refer to the video at this QR code to solve this set of questions.

50. Find the full load secondary current?

    A. 78.70 A

    B. 87.10 A

    C. 80.90 A

    D. 73.55 A

51. What is the emf induced in the secondary?

    A. 318.80 V

    B. 367.11 V

    C. 183.55 V

    D. 370.78 V

52. What is maximum flux in the core?

    A. 41.34 mWb

    B. 119.39 mWb

    C. 44.23 mWb

    D. 81.69 mWb

53. Calculate the regulation of the transformer in which ohmic loss is 1% of the output and the reactance drop is 3% of the voltage when the power factor is 0.8 lag

    A. 2.60 %

    B. 3.10 %

    C. 2.35 %

    D. 1.60 %

54. The open-circuit test result gives 180 V, 85 A, 5 kW measured on the low-voltage side of a transformer. The transformer is rated as 80 kVA, 1000/180 V, 40 Hz and single-phase with 2000 turns on high-voltage side. The core loss component of no load current is approximately.

    A. 27.78 A

    B. 0.03 A

    C. 13.89 A

    D. 36.00 A

55. A 340 KVA, 1100/450 V transformer has primary resistance and secondary resistance of 0.036 Ω and 0.025 Ω. Calculate the copper loss of the transformer, when supplying $(5/7)^{th}$ fraction of full rated load.

    A. 1959 W

    B. 17711 W

    C. data not sufficient

    D. 9036 W

Estimating OC and SC Test Readings

The parameters of a 4.6 KVA, 480/210 V 50 Hz transformer are:

Note: All parameters are referred to the primary side.

Equivalent resistance	=	0.276 Ω
Equivalent reactance	=	0.339 Ω
Core loss resistance	=	693 Ω
Magnetization reactance	=	392 Ω

If a student correctly performs open circuit (OC) test and short circuit (SC) test on the transformer then calculate what the meter readings will be for the above tests. Based on the above data solve following six questions.

Hint: Refer to the video at this QR code to solve this set of questions.

56. As the general practice and from practical convenience point of view, which side of the transformer should the

meters be connected for performing the OC test and SC Test, respectively?

A. HV side, HV side

B. HV side, LV side

C. LV side, HV side

D. LV side, LV side

57. What is the input power taken from the supply during the OC test?

A. 63.6 W

B. 12.2 W

C. 25.3 W

D. 332.5 W

58. What is the input power taken from the supply during the SC test?

A. 132.4 W

B. 332.5 W

C. 4.9 W

D. 25.3 W

59. What is supply voltage to be applied during the SC test?

A. 480 V

B. 0 V

C. 4.2 V

D. 210 V

60. What is current taken from the supply during the OC test?

A. 1.583 A

B. 3.216 A

C. 9.583 A

D. 0.615 A

OC and SC Test on a Transformer

A 220/400 V, 50 Hz transformer when tested gave the following results:
OC test, measurement on LV side:
55 W, 0.7 A, 220 V
SC test, measurement on HV side:
88 W, 12 A, 15 V
Based on the above data determine all the parameters of the equivalent circuit of the transformer and accordingly answer the following four questions.

Hint: Refer to the video at this QR code to solve this set of questions.

61. Determine R_{01}, the equivalent resistance of the transformer in Ω referred to the LV side.

A. 0.611

B. 0.185

C. 880

D. 0.330

62. Determine X_0, the magnetizing reactance of the transformer referring to the LV side.

A. 880

B. 0.330

C. 336

D. 1.090

63. Determine X_{02}, equivalent reactance of the transformer in Ω referred to the HV side.

A. 0.330

B. 1.090

C. 0.611

D. 336

64. Determine R_0, the resistance representing the core losses of the transformer referring to the LV side.

A. 0.611

B. 880

C. 0.185

D. 336

65. A 20 kVA, 1600/240 V, single phase transformer has the following resistance and reactance parameters on primary and secondary sides: $R_1 = 6\ \Omega$, $R_2 = 0.05\ \Omega$, $X_1 = 13\ \Omega$, $X_2 = 0.1\ \Omega$ Find the voltage regulation when transformer is supplying full-load 0.87 pf lagging.

A. 12.308 %

B. -1.131 %

C. -0.170 %

D. 1.846 %

Scan / click the QR code to submit your answers.

* * * All the Best * * *